MIL-M-10670A
15 AUGUST 1952
SUPERSEDING
MIL-M-10670(SigC)
9 NOVEMBER 1950

MILITARY SPECIFICATION

MONITORING SET, TELEPHONE (MONITORING EQUIPMENT AN/PTA-1( ))

This specification has been approved by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 This specification covers one type of small portable, battery operated, audio amplifier equipment intended for use in monitoring telephone transmissions on wire lines. This equipment includes a pickup device which is connected to the input of the amplifier and amplifier output terminations which provides for connection to Headset HS-30-( ), Recording Equipments RC-169-( ) and RC-179-( ) or field wire.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

UU-T-111 —Tape; Paper, Gummed (Sealing and Securing).

MILITARY

JAN-S-28 —Sockets, Electron Tube; and Accessories.

MIL-P-75 —Packaging, Packing, and Container—Marking—Tubes, Electron.

JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

MIL-B-107 — Boxes, Wood, Wirebound (Overseas Type).

JAN-P-108 —Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.

MIL-P-116 — Preservation, Methods of.

JAN-P-117 —Packaging and Packing for Overseas Shipment — Bags, Interior Packaging.

JAN-P-127 —Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.

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MIL–B–131 —Barrier-Materials, Moisture Vaporproof, Flexible.

JAN–T–152 — Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment: General Process for.

MIL–S–901 —Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.

MIL–T–10513 — Tropicalization of Materials Used in Signal Corps Equipment.

MIL–P–11268 — Parts, Materials, and Processes Used in Signal Corps Equipment; General Specification.

U S. ARMY

19–10 — Filler; Interior Packing, Fiberboard, Corrugated, Single and Double Faced, for General Packing Purposes.

72–53 — Finishes (for Ground Signal Equipment).

STANDARDS

MILITARY

JAN–STD–15 —Electrical and Electronic Symbols.

MIL–STD–105— Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

DRAWINGS

SIGNAL CORPS

460–1 —Control Test Sampling Plans for 5% Defective Acceptable Quality Level.

460–2 —Control Test Sampling Plans for 5% Defective Acceptable Quality Level.

SC–D–15800 —Oscilloscope BC–000–( ), Circuit Label (Schematic Diagram).

SC–D–15914 —Moisture Resistance Test Cycle for Ground Signal Equipment.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies)

3. REQUIREMENTS

3.1 Model.—The referenced model shall be understood as the one which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, the equipment shall conform to the following:

3.1.1 *Functions.*—The equipment shall incorporate all functions provided by the model.

3.1.2 *Performance.*—Performance of the equipment shall be at least equal to that of the model. When any performance characteristic of the model is not specified herein, the following shall apply: As soon as the model is received by the contractor, he shall adjust it for optimum performance and prepare to conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance to send a representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

3.2 Preproduction samples.

3.2.1 *Quantities of preproduction samples.* —After award of contract, two preproduction samples of Monitoring Equipment AN/PTA–1( ) shall be furnished for performance test.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be

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accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement (see 3.2.2.1) describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 *Test data.*—The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.2.4 *Reference standards.*—After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.3 Parts, materials and processes; general.—Unless otherwise specified herein, parts, materials and processes shall conform to Specification MIL–P–11268.

3.4 Metals.—The metals used shall be of the proper alloy and hardness necessary to provide the required strength and rigidity. Materials shall be of corrosion-resisting types or shall be suitably processed to resist corrosion. Dissimilar metals used in contact with each other shall be carefully selected in accordance with the requirements of Specification No. 72–53 to prevent electrolytic corrosion in the presence of moisture.

3.5 Insulation material.—Insulation material which will support rapid combustion shall not be used. All insulating material except that used on wire shall be sufficiently rigid and free from heat distortion to maintain its function when the equipment in which it is used is exposed to its maximum specified ambient temperature at 100 percent relative humidity while in operation, storage, or during transportation.

3.6 Ruggedness and reliability.—All parts and details of construction shall have the maximum of ruggedness and reliability under all operating conditions. The equipment shall be so constructed that it will withstand long periods of service and repeated rough handling in transportation, with a minimum of damage or readjustment.

3.7 Temperature and humidity. — The equipment shall be capable of satisfactory operation after storage at any temperature within the ranges from −80° F. to +160° F., at any relative humidity up to 100 percent, and at any altitude up to 35,000 feet. The equipment shall be capable of operation in any external ambient temperature between the limits of −65° F. and +150° F., and all ranges of humidity but not necessarily meeting full specification requirements. Full specification requirements shall be met in any external ambient temperature within the range −40° F. to +131° F. at any relative humidity up to 100 percent and at altitudes up to 10,000 feet. All items or parts entering into the construction of any unit of the complete equipment shall be capable of operating continuously, without damage, at any external ambient temperature specified herein.

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3.8 Vibration, bounce, and shock.

3.8.1 *Vibration.* — Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed.

3.8.2 *Bounce and shock.*—The equipment shall meet the tests of 4.18.2 and 4.18.3.

3.9 Circuit diagrams. — Circuit diagrams shall be furnished as required to facilitate installation, operation and maintenance of the equipment The diagrams shall be in the form of labels meeting the general requirements of Specification MIL–P–11268.

3.9.1 *Schematic diagrams.*—Schematic diagrams shall be suitable for use by maintenance personnel and shall be mounted so far as practicable on inside surfaces where they will not be exposed to abrasion or the elements but will be readily accessible to maintenance personnel. Graphical symbols on schematic diagrams shall conform to Military Standard JAN–STD–15. Schematic diagrams shall conform to the example shown on Drawing SC–D–15800.

3.10 Tropicalization. — The assembled equipment shall be treated in accordance with Specification JAN–T–152. Before being assembled into the equipment, materials and parts shall be treated in accordance with Specification MIL–T–10513.

3.11 Running spare parts.—Running spare parts shall be furnished as specified (see 6.2). Running spare parts shall be identical to corresponding integral parts of the equipment furnished on order.

3.12 Construction. — The equipment covered by this specification shall be constructed in accordance with the model, except as otherwise specified herein, and shall meet all requirements and tests of this specification.

3.13 Physical requirements.

3.13.1 *Amplifier unit.*

3.13.1.1 *Weight.*—The weight of the amplifier unit complete with batteries and with items stored in the cover shall not exceed 22 pounds. The combined weights of the canvas bag, three Ground Rods GP–106/G and 500 feet of Wire WD–1/TT on reel shall not exceed 15 pounds.

3.13.1.2 *Materials.*

Case and cover: Aluminum, 14 B & S gage, heat treated to 61 ST. Hardness.

Magnesium sheet, 12 B & S gage with Dowmetal alloy J–1a, or equal.

Chassis: 0.064 52S ¼ H Aluminum. Magnesium same as for case.

Panel: 52S ¼ H Aluminum or Dowmetal alloy J–1h, or equal.

Core of pickup transformer: Mu metal. Allegheny Ludlum Steel Co., or equal.

3.13.2 *Vacuum tube sockets* —The vacuum tube sockets shall be JAN Type TS102P01 in accordance with Specification JAN–S–28.

3.13.3 *Watertightness.*—The case for the amplifier unit shall be completely watertight both when open for operation and when closed for transportation. The lid of this case shall provide weather protection to the input and output terminals when the line wires and headset leads are connected for operation.

3.14 Electrical characteristics.

3.14.1 *Amplifier unit.*

3.14.1.1 *Gain frequency characteristics.*—The amplifier shall provide 90 db gain at 1,000 cps and not less than 85 db gain throughout the frequency range from 200 cps to 8,000 cps. The gain frequency curve shall be relatively smooth and free from humps. The 15,000 cps response shall be at least 10 db lower than the response at 1,000 cps.

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3.14.1.2 *Filters.*—The high pass filter shall offer a loss of 30 db or more at frequencies below 300 cps and not more than 3 db loss at 500 cps. The low pass filter shall offer a loss of 30 db or more at frequencies above 3,000 cps and not more than 3 db loss at 2,000 cps.

3.14.1.3 *Gain control.*—The gain control shall consist of a 24 step attenuator with the first two steps equivalent to 10 db each, the top step equivalent to 5 db, and the intervening steps equivalent to 3 db each. The accuracy of these steps shall be ± 0.5 db.

3.14.1.4 *Power output.*—The amplifier unit shall be capable of producing an output power of plus 10 dbm in a 600 ohm load at the output terminals at any frequency between 200 cps and 8,000 cps. At this output power, the harmonic distortion shall not exceed 7 percent.

3.14.1.5 *Impedance.*—The output impedance of the amplifier shall be 600 ohms plus or minus 10 percent at 1,000 cps. The input impedance shall consist of the following steps at 1,000 cps: 20 ohms, 150 ohms, 600 ohms, 1,000 ohms, 2,000 ohms, 3,500 ohms, 5,000 ohms and 500,000 ohms. These impedance values shall be within plus or minus 10 percent when both filters are out of the circuit, plus or minus 20 percent when either the low pass filter or the high-pass filter is in the circuit, and plus or minus 25 percent when both filters are in the circuit. The impedance balance at the output shall be at least 60 db. The impedance balance at the input shall be at least 60 db.

3.14.1.6 *Bridging loss.*—The bridging loss caused by the direct connection of Monitoring Equipment AN/PTA–1( ) to a single conductor ground return line or a two wire metallic line shall be less than 0.1 db.

3.14.1.7 *Noise level.*—The noise output of the amplifier units shall be not more than 44 db RN when measured by Transmission Measuring Set TS–559( )/FT with normal line weighting.

3.14.1.8 *Battery drain.*—The "B" current shall not exceed 5 milliamperes at 90 volts. The "A" current shall not exceed 50 milliamperes at 6 volts.

3.14.2 *Induction pickup transformer.*—The induction pickup transformer shall supply adequate signal strength to operate the amplifier at 0 dbm output with at least 5 db reserve gain, when clamped on either wire of a two wire circuit or on the single wire of a ground return circuit when that circuit is energized with a 1,000 cps signal at a power level of minus 30 dbm at the point of signal pickup.

3.14.2.1 *Frequency response.*—The induction pickup transformer shall have a frequency response such that when connected to the amplifier it will not cause a variation of more than plus or minus 10 db from the 1,000 cps value of the amplifier output over the frequency range from 300 cps to 8,000 cps.

3.15 Workmanship.—Components, subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection, general. — Inspection shall consist of group A and B inspection as listed in table I and as specified in 4.1.1 and 4.1.2 (see 6.2). Group B inspection shall be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, this order of inspection may be varied when the Government inspector considers it more practical for administrative reasons to select samples for group B inspection. Samples for inspection shall be selected at random by the Government inspector.

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4.1.1 *Group A inspection.*—At the discretion of the Government inspector, group A inspection shall be conducted by either or both of the following methods:

(a) 100 percent inspection.

(b) Statistical sampling in accordance with Military Standard MIL–STD–105. AQL's (acceptable quality levels) shall be as shown in Table I or the contract.

Order of inspection: Any order which is satisfactory to the Government inspector may be used:

4.1.2 *Group B inspection.*—The AQL's for group B inspection shall be as shown in table I or the contract. Sampling plans for specified AQL's shall conform to Drawings 460–1 and 460–2.

4.1.2.1 *Rejection.*—If the number of defective sample units from a lot exceeds the specified acceptance number, the lot shall be rejected and the contractor shall take immediate action to correct the cause of failure.

4.1.2.2 *Order of inspection.*—Group B inspection shall be performed in the order shown in table I.

4.1.3 *Disposition of rejected product.*—If a defective unit or lot is resubmitted for inspection, the unit or lot shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government inspector when the unit or lot is resubmitted and shall become the property of the Government.

4.1.3.1 *Groups A and B.*—Disposition of product rejected during group A or group B inspection shall be in accordance with Military Standard MIL–STD–105.

4.2 **Visual and mechanical inspection.**—Parts and equipment shall be inspected for

TABLE I.—*Inspection Procedures.*

Inspection or Test	Requirement Paragraph	Test Paragraph	Preproduction	Group A AQL DPHU*	Group B AQL
Visual and mechanical, major	--	4.2	X	1.5%	--
Visual and mechanical, minor	--	4.2	X	6.5%	--
Dielectric	--	4.3	X		--
Gain control	3.14.1.3	4.7	X	1.5%	--
Operational	--	4.4	X	--	--
Gain frequency	3.14.1.1	4.5	X	--	
Filters	3.14.1.2	4.6	X	--	
Power output	3.14.1.4	4.8	X	--	
Output impedance	3.14.1.5	4.9	X	--	
Input impedance	3.14.1.5	4.10	X	--	
Impedance balance at output	3.14.1.5	4.11	X	--	
Impedance balance at input	3.14.1.5	4.12	X	--	5%
Bridging loss	3.14.1.6	4.13	X	--	
Noise	3.14.1.7	4.14	X	--	
Battery drain	3.14.1.8	4.15	X	--	
Signal strength and reserve gain with induction pickup transformer	3.14.2	4.16	X	--	
Frequency response with pickup transformer	3.14.2.1	4.17	X		

* DPHU—Defects per hundred units.

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workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc., shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with definitions of Military Standard MIL–STD–105.

4.3 **Dielectric test.**—When practicable (provided that no unavoidable unsoldering of connections is required and components are not overrated by the applied voltage), the dielectric strength of all circuits on each equipment shall be tested to withstand 400 volts or two times the maximum peak operating voltage whichever is greater. The test voltage shall be d-c or peak a-c of the same value, and shall be applied for 10 seconds between wiring and adjacent nonconductors and ground. All tubes shall be removed and the protectors shall be disconnected from the circuit before the test voltage is applied.

4.4 **Operational test.**—The induction pickup transformer shall be clamped on one wire of a two wire line which is approximately six feet in length, has a signal generator at one end, and is terminated by a 600 ohm resistor at the other end. The signal generator shall be adjusted to 1,000 cps with an output power level of minus 30 dbm. The signal strength of the amplifier output shall be 0 dbm when the volume control is set at 85 and when the filter switch is varied through each of its positions. The test shall be performed with internal batteries and shall be repeated with external batteries. The noise in the amplifier unit shall be checked with the headset that is to be supplied with the sample under test.

4.4.1 Any continuity test which is not covered in 4.4 shall be made on all electrical circuits. This test shall be made with the switches in each combination of operating positions.

4.5 **Gain frequency.**—The gain frequency characteristics between the 600 ohm input circuit and the output circuit terminated by a 600 ohm resistor shall be measured over the frequency range from 200 cps to 1,5000 cps in each filter condition. The results of these measurements shall meet the requirements of 3.14.1.1.

4.6 **Filters.**—The filters shall be tested as an integral part of the amplifier unit by making gain frequency runs under filtered conditions and comparing them with the nonfiltered gain characteristics. The characteristics of the filters shall be such as to meet the requirements of 3.14.1.2.

4.7 **Gain control.**—The steps of the gain control switch shall be tested at a frequency of 1,000 cps. The gain control shall meet the requirements of 3.14.1.3.

4.8 **Power output.**—With the power output adjusted to + 10 dbm, the harmonic distortion shall be measured at the following frequencies: 200, 1,000, 8,000 cps for compliance with 3.14.1.4.

4.9 **Output impedance.**—The output impedance shall be measured at 1,000 cps for each filter condition. The output impedance shall meet the requirements of 3.14.1.5.

4.10 **Input impedance.**—The input impedance at 1,000 c.p.s. shall be measured at each setting of the input impedance switch and for each filter condition. The input impedance shall meet the requirements of 3.14.1.5.

4.11 **Impedance balance at output.**—Measurements shall be made in accordance with the test circuits shown in figures 1 and 2. In each case, the oscillator and attenuator shall be so adjusted that with maximum gain in the amplifier unit the output under test

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conditions of figure 1 shall be +10 dbm. The difference between the results obtained under test conditions of figures 1 and 2 shall meet the requirements of 3.14.1.5.

4.12 **Impedance balance at input.**—The input impedance switch shall be set at the 600 ohm position. The difference between the gain of the amplifier as determined by the test condition specified in figure 1 and the results obtained under the test condition shown in figure 3 shall be in accordance with the requirements of 3.14.1.5.

4.13 **Bridging loss.** — The bridging loss caused by direct connection of the monitoring equipment to a single conductor ground return line or a two wire metallic line shall be measured and shall be within the requirements of 3.14.1.6.

4.14 **Noise.**—Overall noise shall be measured on a Transmission Measuring Set TS–559( )/FT or equal, with normal line weighting and shall be within requirements of 3.14.1.7.

4.15 **Battery drain.**—The battery drain of both the A and B batteries shall be measured. The current shall not exceed the values specified in 3.14.1.8.

4.16 **Signal strength and reserve gain with induction pickup transformer.** — The signal strength and reserve gain of the sample with induction pickup transformer shall be tested for compliance with 3.14.2.

4.17 **Frequency response with pickup transformer.**—The frequency response shall be tested over the frequency range from 300 cps to 8,000 cps and compared to the output of the amplifier at 1,000 cps. The results of these measurements shall meet requirements of 3.14.2.1.

4.18 **Preproduction tests.**—The service conditions tests will be performed on the preproduction samples at a Government laboratory Following completion of the service conditions tests, the performance of the equipment shall be checked and comparison of performance characteristics shall be made with initial test results. Without other than normal adjustment, the operation of the components shall be satisfactory to the contracting officer.

4.18.1 *Vibration test.*—The equipment shall be tested for resonance frequencies below 55 cycles per second to determine conformance with 3.8.1.

4.18.2 *Bounce test.*—The equipment, in its transit case, shall be placed on the table of the package tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ± 1 percent, for a total of 3 hours. During each ½-hour period of the test, the transit case (with contained equipment) shall rest on a different face. Any evidence of degradation of performance below limits specified herein shall be cause for rejection.

4.18.3 *Shock test.*—The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" as shown in Specification MIL–S–901.

4.18.3.1 The equipment, in its transit case and bag, shall be secured to test plate of the shock-testing machine in the position that the equipment will occupy during operation. Any shock-absorbing devices shall also be in place. The equipment shall be securely fastened to the test plate with not less than 4 bolts. The test shall consist of a total of 9 blows as follows:

One each 1-foot, 3-foot and 5-foot back blow

One each 1-foot, 3-foot and 5-foot side blow

One each 1-foot, 3-foot and 5-foot top blow

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TEST CIRCUITS FOR
IMPEDANCE BALANCE MEASUREMENTS

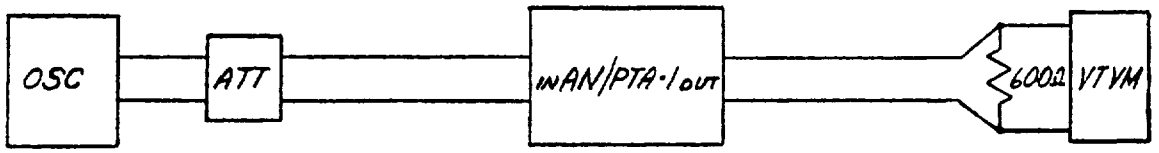


FIGURE 1

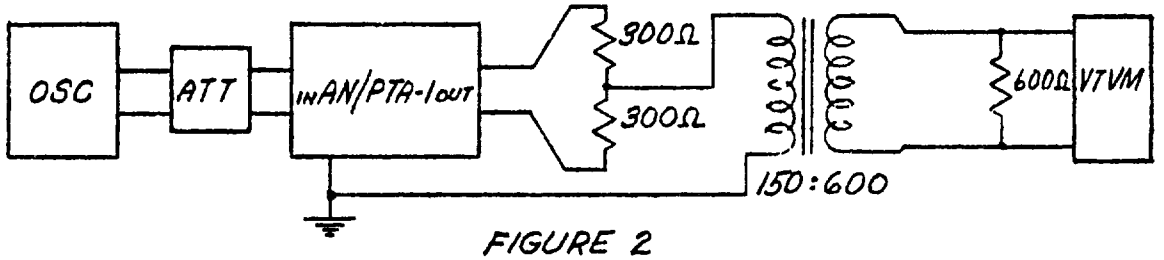


FIGURE 2

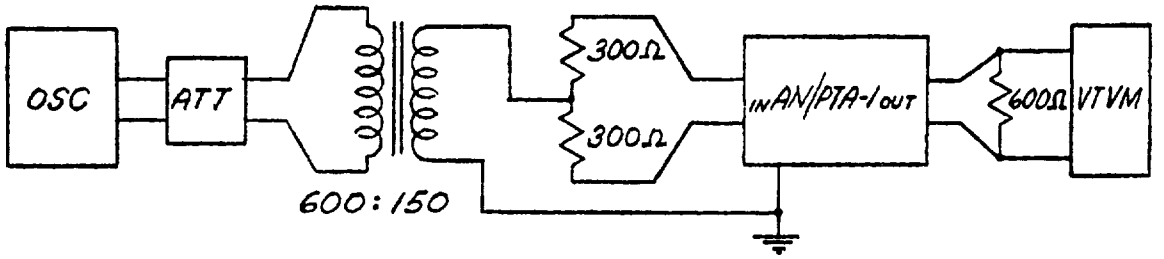


FIGURE 3

OSC. SHALL BE WESTERN ELECTRIC CO. 19C OR EQUAL.
ATT. SHALL BE DAVEN CO. TYPE H-693 OR EQUAL.
VTVM SHALL BE BATTERY OPERATED BALLANTINE MODEL 300A OR EQUAL.
ACCURACY OF RESISTORS SHALL BE ±0.1%.

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There shall be no evidence of collision between any parts of the equipment during this test. There shall be no breakage, permanent deformation, or loosening of parts.

4.18.4 *Low temperature test.*—The components covered by this specification shall be maintained for 3 days in a chamber at an ambient temperature of —80° F. The temperature shall then be raised to —40° F., and stabilized. The components shall be capable of meeting all specification requirements at this temperature.

4.18.5 *High temperature test.*—The components covered by this specification shall be maintained in a chamber at an ambient temperature of + 160° F. for three days. The temperature shall then be lowered to + 131° F. and stabilized. The equipment shall be capable of meeting all specification requirements at this temperature.

4.18.6 *Moisture-resistance test for equipment.*

4.18.6.1 *Test conditions.*

(a) Do not move components from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the components. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.18.6.2 *Test procedure.*—The components will be tested as follows:

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Measure the performance of the samples in accordance with 4.5 through 4.17 inclusive. Readjust, if necessary, to meet specification requirements.

(c) Subject the samples to continuous cycling for a total of 5 of the 48 hour cycles shown on Drawing SC–D–15914. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC–D–15914. Measure their performance in accordance with 4.5 through 4.17 inclusive, during the periods indicated on the drawing.

4.18.6.3 *Performance.*—There shall be no degradation in the performance of the equipment from that obtained prior to the five humidity cycles. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or the binding of rotating parts.

4.18.7 *Immersion test.* — The immersion test of the amplifier unit shall be conducted as follows:

(a) A source of dry air or nitrogen with necessary gages and hoses shall be provided for the purpose of inserting air into the equipment under test.

(b) Connect a pressure hose to the valve or fitting that has been inserted in the tapped hole in the panel, and increase dry air or nitrogen pressure until the internal pressure of the equipment is 3 pounds per square inch.

(c) Close valve to equipment and remove air hose.

(d) Immerse unit for at least one minute in a glass water tank filled with water at room temperature. The cover of the amplifier unit shall be open and the top of the panel shall be at least one foot below the surface of the water. No bubbles shall appear in the water during the time that the unit is immersed.

(e) Remove unit from tank and measure the pressure in the unit. The pres-

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sure reading must be the same as the reading taken before the unit was immersed.

(f) In the event that a unit fails in the immersion test, it shall be fully retested and shall meet this test before it will be considered acceptable.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *For domestic shipment.*—Monitoring Equipment AN/PTA–1 ( ) shall be preserved and packaged in accordance with method III of Specification MIL–P–116. The equipment shall be adequately cushioned, blocked, and braced using suitable materials and containers as required, in such a manner as to afford maximum protection from hazards normally encountered in storage and transit.

5.1.2 *For export shipment.*—Preserve and package Monitoring Equipment AN/PTA–1( ) in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116, and further described in the following paragraphs:

5.1.2.1 *Technical manual.*—Package each technical manual method IC–3, as follows: Inclose each technical manual within a close-fitting type II, class b bag, conforming to Specification JAN–P–117. Properly seal closure.

5.1.2.2 *Tubes.*—Remove all operating tubes over 4 inches in length not securely clamped in place and package together with spare tubes; determined in quantity and number from the bid request and contract, individually in accordance with the applicable provisions of Specification MIL–P–75.

5.1.2.3 *Bag containing Ground Rods GP–106/G and reel.*—Package each item method III as follows: Cushion each ground rod and reel as required by wrapping each item with flexible corrugated paper. Secure wrapping with gummed Kraft tape conforming with Specification UU–T–111. Place cushioned ground rods and reel within bag. Fill all voids of bag to prevent movement. Close cover and securing fastenings.

5.1.2.4 *Consolidated package.*—Consolidate the items, packaged as specified in 5.1.2.2 and 5.1.2.3, together within a suitable style and grade fiberboard box, conforming to Specification JAN–P–108. Fill all voids with flexible corrugated paper to prevent movement. Seal entire box closure with water-resistant pressure sensitive tape conforming to Specification JAN–P–127.

5.1.2.5 *Amplifier unit in immersion test case.*—Package each amplifier in case method IC–5, as follows: Cushion each amplifier in case on all surfaces with cells or pads or both, fabricated of corrugated fiberboard, conforming to Specification 19–10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned case within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN–P–108. Seal entire closure with water-resistant pressure sensitive tape.

5.2 Packing.

5.2.1 *For domestic shipment.*—Monitoring Equipment AN/PTA–1( ), packaged as specified in 5.1.1, shall be packed in substantial containers of the type, size and kind commonly used for the purpose, in such a manner as to insure acceptance at and safe delivery to the designated point by common or other carrier at the lowest rate.

5.2.2 *For export shipment.*—The Monitoring Equipment AN/PTA–1( ) packaged as specified in 5.1.2.5, consolidated package as specified in 5.1.2.4, together shall be packed in a wood cleated plywood, nailed wood or wire bound wood box conforming to the requirements of Specifications JAN–P–105, JAN–P–106 and MIL–B–107 respectively.

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Fabricate the boxes to fit contents snugly. Place technical manual packaged as specified in 5.1.2.1 between packaged item and lid of shipping container. Close and strap the shipping container as specified in the appendix of the applicable box specification (Shipments consigned to depots shall not require strapping).

5.3 Marking for shipment.—Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of Military Standard MIL–STD–129.

5.4 Preservation and packaging tests.

5.4.1 Items preserved and packaged in section 5 shall be capable of passing test requirements as applicable in Specification MIL–P–116.

5.4.2 Prior to packaging (and packing) the equipment in quantity, a pilot pack shall be submitted to the Government inspector for approval to determine conformance with requirements of this specification. Approval of the pilot pack shall not relieve the contractor of his obligation to process, package and pack the equipment in compliance with the requirements of this specification.

6. NOTES

6.1 Intended use.—Monitoring Equipment AN/PTA–1( ) is intended for use as a lightweight battery operated tactical military equipment for the field interception of enemy wire line voice frequency messages without detection.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether domestic or oversea shipment is required.
(c) Samples required (See 3.2).
(d) Marking and shipping of samples.
(e) Place of final inspection (see 4.1).
(f) Acceptable quality level (see 4.1.1-(b), 4.1.2, and table I).
(g) Facsimiles required.
(h) Running spares required (see 3.11).

6.3 Nomenclature.—The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example AN/PTA–1W. As soon as possible after the award of contract, the contractor should apply to the contracting officer for such information.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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MIL-M-10670A MONITORING SET, TELEPHONE AN/PTA-1()

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